

Implications of $g - 2$ and NA62 result for the 3-3-1 models.

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Dead or alive? Implications of the muon anomalous magnetic moment
for 3-3-1 models

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**Rare kaon decay to missing energy: Implications
of the NA62 result for a Z' model**

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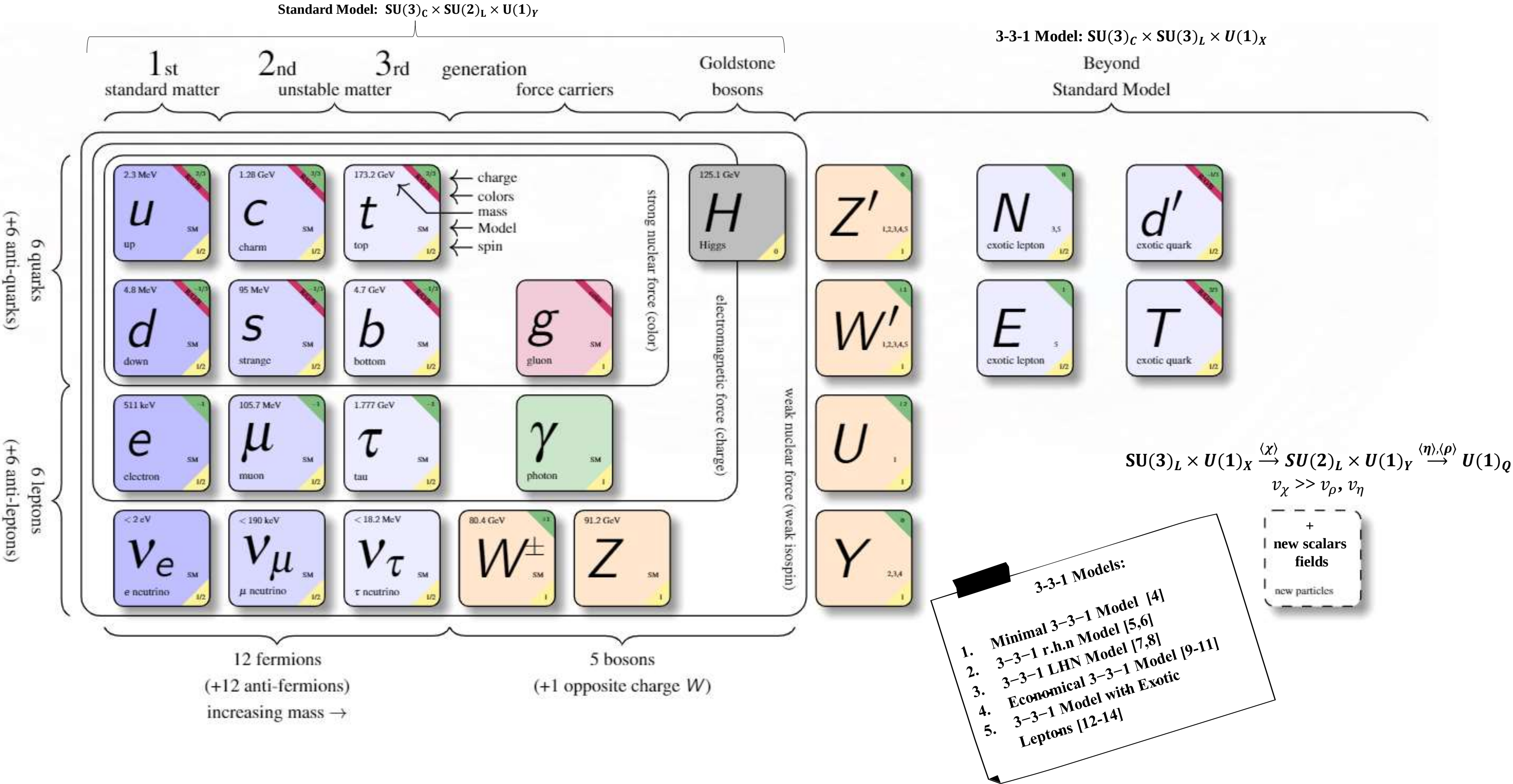
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Muon Anomalous Magnetic Moment (a_μ)

Rare kaon decay $K^+ \rightarrow \pi^+ \nu \bar{\nu}$

3-3-1 Models

— PART 1



Muon Anomalous Magnetic Moment (a_μ)

Through quantum loop effects: $a_\mu^{\text{SM}} \equiv \frac{g_\mu - 2}{2}$

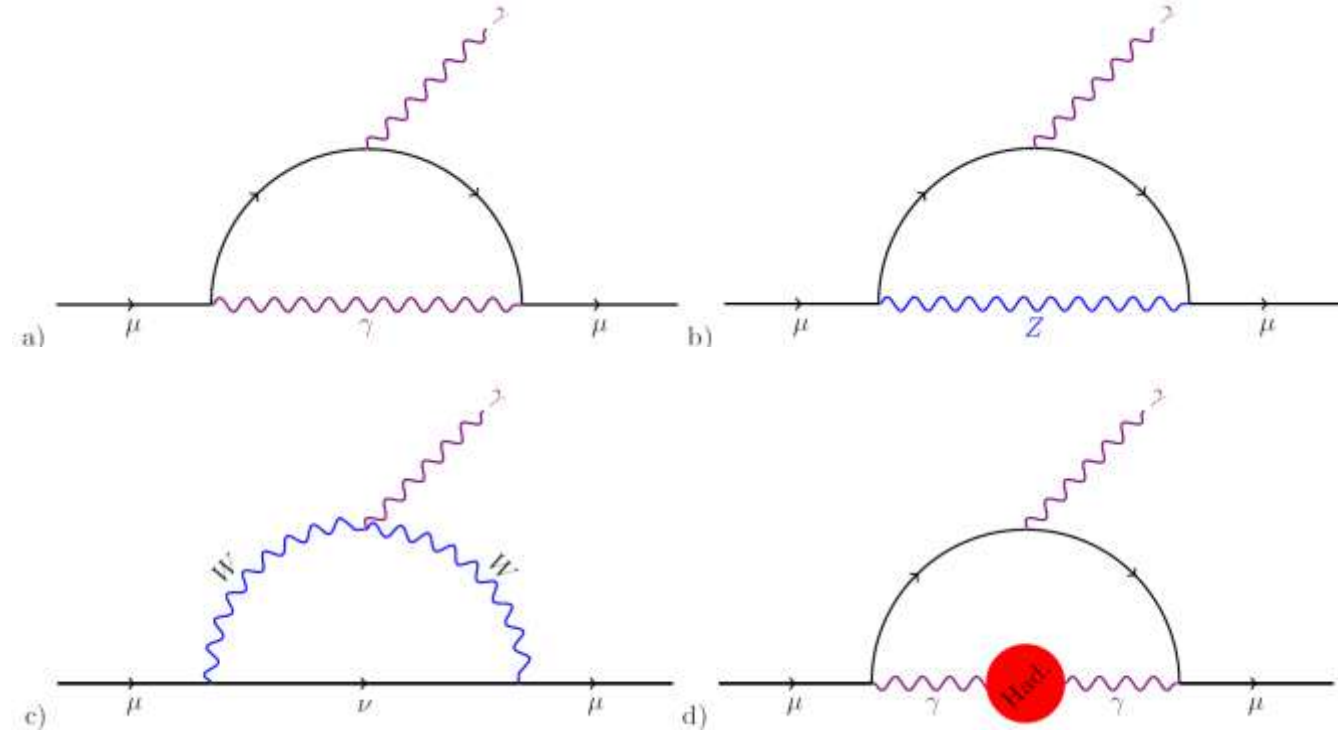


Figure 1: Feynman diagrams of the corrections to a_μ^{SM} on SM interactions: a) first order QED, b) and c) lowest-order weak, and d) lowest-order hadronic effects.

$$\Delta a_\mu = a_\mu^{\text{exp}} - a_\mu^{\text{SM}} \text{ limited by: } \begin{cases} \text{Experimental uncertain: FERMILAB and J-PARC} \\ \text{Theoretical uncertain: hadronic effects} \end{cases}$$

$$\Delta a_\mu = (251 \pm 59) \times 10^{-11} (4.2\sigma) - (\text{FERMILAB} - 2021) [18]$$

We have implemented as current discrepancy:

$$\Delta a_\mu = (261 \pm 78) \times 10^{-11} (3.3\sigma) [1, 2] - (\text{Brookhaven National Lab} - 2009)$$

We have imposed a projected discrepancy: $\Delta a_\mu = (261 \pm 34) \times 10^{-11} (5\sigma)$

Rare kaon decay $K^+ \rightarrow \pi^+ \nu \bar{\nu}$

$$BR(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = (8.4 \pm 1.0) \times 10^{-11} \text{ (SM)}$$

$$BR(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = (11^{+4.0}_{-3.5}) \times 10^{-11} \text{ (NA62 -2016, 2017 and 2018)}$$

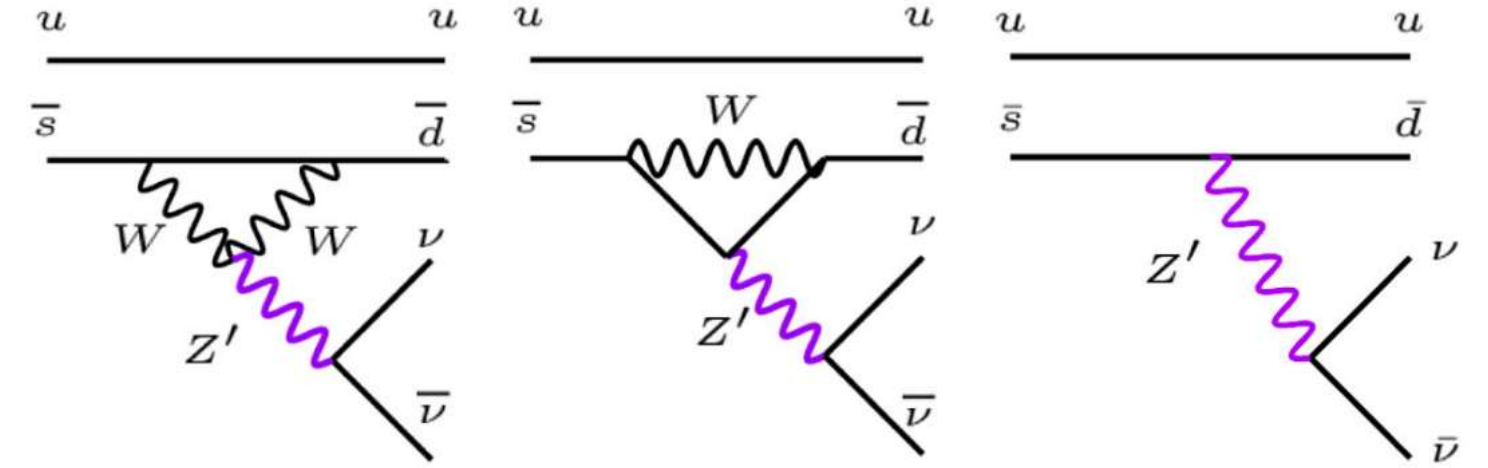


Figure 2: Feynman diagrams that illustrate how vector mediators can contribute to the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$. The first diagram requires Z' coupling to neutrinos, the second further requires couplings to the top quark, whereas in the third case the Z' boson mediates FCNC at tree level, which is the main focus of this work.

The quark fields in the following standard rotation are

$$\begin{pmatrix} u \\ c \\ t \end{pmatrix}_L = V_L^u \begin{pmatrix} u' \\ c' \\ t' \end{pmatrix}_L, \quad \begin{pmatrix} d \\ s \\ b \end{pmatrix}_L = V_L^d \begin{pmatrix} d' \\ s' \\ b' \end{pmatrix}_L$$

Where V_L^u and V_L^d are the 3×3 unitary matrices such that for the Cabibbo-Kobayashi-Maskawa (CKM) matrix we have $V_{\text{CKM}} \equiv V_L^{u\dagger} V_L^d$.

V_L^d matriz [19]:

$$V_L^d = \begin{pmatrix} \tilde{c}_{12}\tilde{c}_{13} & \tilde{s}_{12}\tilde{c}_{23}e^{i\delta_3} - \tilde{c}_{12}\tilde{s}_{13}\tilde{s}_{23}e^{i(\delta_1-\delta_2)} & \tilde{c}_{12}\tilde{c}_{23}\tilde{s}_{13}e^{i\delta_1} + \tilde{s}_{12}\tilde{s}_{23}e^{i(\delta_2+\delta_3)} \\ -\tilde{c}_{13}\tilde{s}_{12}e^{-i\delta_3} & \tilde{c}_{12}\tilde{c}_{23} + \tilde{s}_{12}\tilde{s}_{13}\tilde{s}_{23}e^{i(\delta_1-\delta_2-\delta_3)} & -\tilde{s}_{12}\tilde{s}_{13}\tilde{c}_{23}e^{i(\delta_1-\delta_3)} - \tilde{c}_{12}\tilde{s}_{23}e^{i\delta_2} \\ -\tilde{s}_{13}e^{-i\delta_1} & -\tilde{c}_{13}\tilde{s}_{23}e^{-i\delta_2} & \tilde{c}_{13}\tilde{c}_{23} \end{pmatrix}$$

where $\tilde{s}_{ij} = \sin \tilde{\theta}_{ij}$, $\tilde{c}_{ij} = \cos \tilde{\theta}_{ij}$, and δ_i are the phases, with $i, j = 1, 2, 3$.

3-3-1 r.h.n

$d - s - Z'$ coupling :

$$\mathcal{L}_{Z'} \supset \Delta_L^{sd} (\bar{s}_L \gamma^\mu d_L) Z'_\mu + \dots, \quad \text{with} \quad \Delta_L^{sd} = \frac{g C_W}{\sqrt{3-4S_W^2}} V_{L32}^* V_{L31}.$$

$$V_{L31}^d V_{L32}^{d*} = -\tilde{s}_{13}\tilde{c}_{13}\tilde{s}_{23}e^{-i(\delta_1-\delta_2)} \equiv |V_{L32}^{d*} V_{L31}^d| e^{-i\delta}$$

Analogously for the neutrino- Z' coupling:

$$\Delta_L^{\nu\bar{\nu}} = \frac{g}{2C_W} \frac{1 - 2S_W^2}{\sqrt{3 - 4S_W^2}}$$

With $C_W = \cos \theta_W$, $S_W = \sin \theta_W$. θ_W is the electroweak angle.

Contributions to $g-2$

Kaon Decays

— PART 2

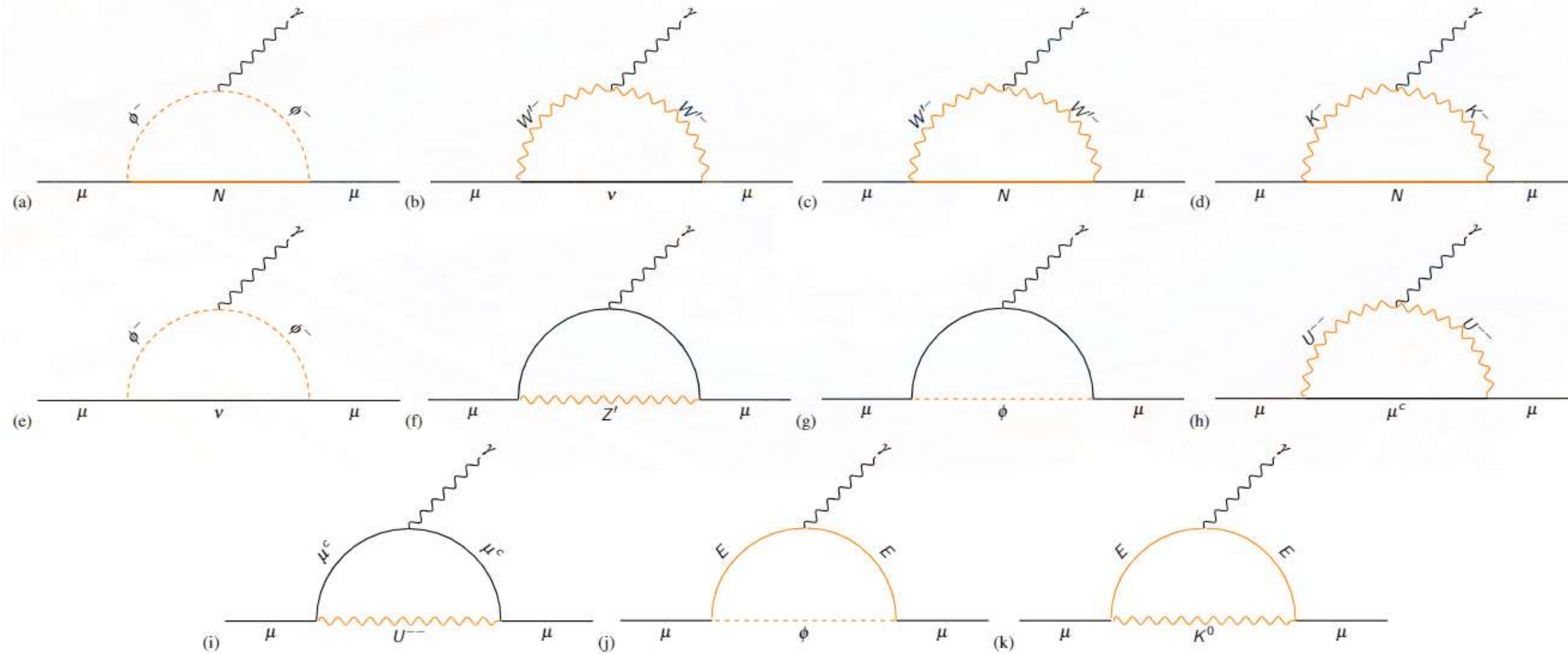


Figure 3: Feynman diagrams contribute to $g - 2$ in the 3-3-1 models investigated in this work.

We make our Mathematica numerical codes of the **analytical expressions for the corrections to $g_\mu - 2$** corresponding to the **3-3-1 Models** available at: <https://github.com/Yoxara/Implications-of-the-Muon-Anomalous-Magnetic-Moment-for-3-3-1-Models>

Kaon Decays

$$\begin{array}{ccc}
 K^+ \rightarrow \pi^+ \nu \bar{\nu} & \text{and} & K_L \rightarrow \pi^0 \nu \bar{\nu} \\
 \downarrow & & \downarrow \\
 \text{conserves CP} & & \text{violates CP}
 \end{array}$$

$$\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = \kappa_+ \left[\left(\frac{\text{Im } X_{\text{eff}}}{\lambda^5} \right)^2 + \left(\frac{\text{Re } X_{\text{eff}}}{\lambda^5} - \bar{P}_c(X) \right)^2 \right]$$

$$\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = \kappa_L \left(\frac{\text{Im } X_{\text{eff}}}{\lambda^5} \right)^2$$

$$\kappa_+ = (5.21 \pm 0.025) \times 10^{-11} \left(\frac{\lambda}{0.2252} \right)^8, \quad \kappa_L = (2.25 \pm 0.01) \times 10^{-10} \left(\frac{\lambda}{0.2252} \right)^8, \quad \lambda \text{ is the}$$

Cabibbo angle, $P_c(X)$ summarizes the charm contribution

$$\bar{P}_c(X) = \left(1 - \frac{\lambda^2}{2} \right) P_c(X), \quad \text{with} \quad P_c(X) = (0.42 \pm 0.03) \left(\frac{0.2252}{\lambda} \right)^4$$

X_{eff} describes the contribution of short distance physics: $X_{\text{eff}} = V_{ts}^* V_{td} X_L(K)$,

$$X_L(K) = \eta_X X_0(x_t) + \frac{\Delta_L^{\nu\bar{\nu}}}{g_{\text{SM}}^2 m_{Z'}^2} \frac{\Delta_L^{sd}}{V_{ts}^* V_{td}} \quad \text{and} \quad g_{\text{SM}}^2 = 4 \frac{m_W^2 G_F^2}{2\pi^2} = 1.78137 \times 10^{-7} \text{GeV}^{-2}$$

$$\eta_X = 0.994, \text{ and } X_0(x_t) = \frac{x_t}{8} \left[\frac{x_t+2}{x_t-1} + \frac{3x_t-6}{(x_t-1)^2} \ln x_t \right] \text{ with } x_t = m_t^2/m_W^2.$$

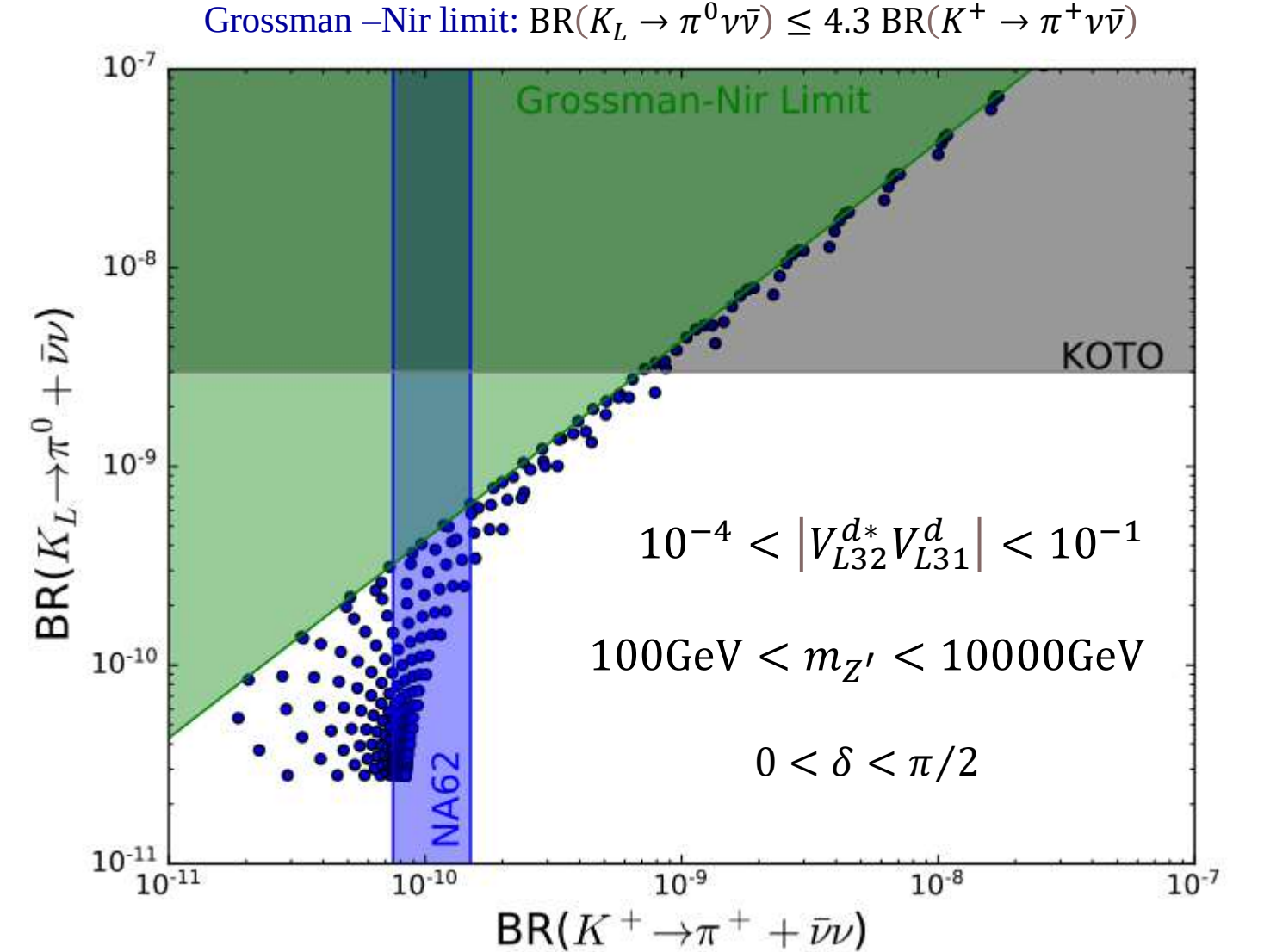


Figure 4: $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu})$ versus $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu})$; the blue dots were generated varying the $m_{Z'}$, the module $|V_{L32}^{d*} V_{L31}^d|$ and the δ phase for the **3-3-1 r.h.n.** model. We combined three different limits, the theoretical Grossman-Nir limit (green region), the NA62 bounds (blue region) and the exclusion region from KOTO (gray region).

Results

— PART 3

Model	LHC-13TeV	g-2 current	g-2 projected
Minimal 3-3-1	$M_{Z'} > 3.7 \text{ TeV}$ [16] $M_{W'} > 3.2 \text{ TeV}$ [16]	$M_{Z'} > 434.5 \text{ GeV}$ $M_{W'} > 646 \text{ GeV}$	$M_{Z'} > 632 \text{ GeV}$ $M_{W'} > 996.1 \text{ GeV}$
3-3-1 r.h.n	$*M_{Z'} > 2.64 \text{ TeV}$ [3] —	$M_{Z'} > 158 \text{ GeV}$ $M_{W'} > 133 \text{ GeV}$	$M_{Z'} > 276.5 \text{ GeV}$ $M_{W'} > 239 \text{ GeV}$
3-3-1 LHN for $M_N = 1 \text{ GeV}$	$*M_{Z'} > 2 \text{ TeV}$ [3] —	$M_{Z'} > 160 \text{ GeV}$ $M_{W'} > 134.3 \text{ GeV}$	$M_{Z'} > 285 \text{ GeV}$ $M_{W'} > 238.3 \text{ GeV}$
3-3-1 LHN for $M_N = 100 \text{ GeV}$	$*M_{Z'} > 2 \text{ TeV}$ [3] —	$M_{Z'} > 136.7 \text{ GeV}$ $M_{W'} > 114.2 \text{ GeV}$	$M_{Z'} > 276.5 \text{ GeV}$ $M_{W'} > 231 \text{ GeV}$
Economical 3-3-1	$*M_{Z'} > 2.64 \text{ TeV}$ [3] —	$M_{Z'} > 59.3 \text{ GeV}$ $M_{W'} > 49.5 \text{ GeV}$	$M_{Z'} > 271.4 \text{ GeV}$ $M_{W'} > 226.7 \text{ GeV}$
3-3-1 exotic leptons for $m_N = 10 \text{ GeV}$, $m_E = 150 \text{ GeV}$	$*M_{Z'} > 2.91 \text{ TeV}$ [17] —	$M_{Z'} > 429 \text{ GeV}$ $M_{W'} > 359 \text{ GeV}$	$M_{Z'} > 693 \text{ GeV}$ $M_{W'} > 579.6 \text{ GeV}$
3-3-1 exotic leptons for $m_N = 10 \text{ GeV}$, $m_E = 150 \text{ GeV}$	$*M_{Z'} > 2.91 \text{ TeV}$ [17] —	$M_{Z'} > 369 \text{ GeV}$ $M_{W'} > 309.1 \text{ GeV}$	$M_{Z'} > 600 \text{ GeV}$ $M_{W'} > 501.4 \text{ GeV}$

TABLE I. Summary of the lower bounds based on our calculations. For comparison, we include the LHC bounds at 13 TeV center-of-mass energy. These LHC bounds are based on either 36 fb or 139 fb of data. The lower mass bounds on the Z' and W' bosons from the 3-3-1 r.h.n model are also applicable to the Economical 3-3-1 model as they have the same interactions. Conversely, the lower mass constraints we found are robust.

Extended version of the 3-3-1 LHN Model by an inert scalar triplet

The inert scalar triplet allows us to include $\mathcal{L} \supset y_{ab} \bar{f}_a \phi e_{bR}$, taking $y_{22} = 1$. Such scalar triplet gets a mass from the quartic coupling in the scalar potential $(\lambda \phi^\dagger \phi \chi^\dagger \chi)$, after the scalar triplet χ acquires a vev.

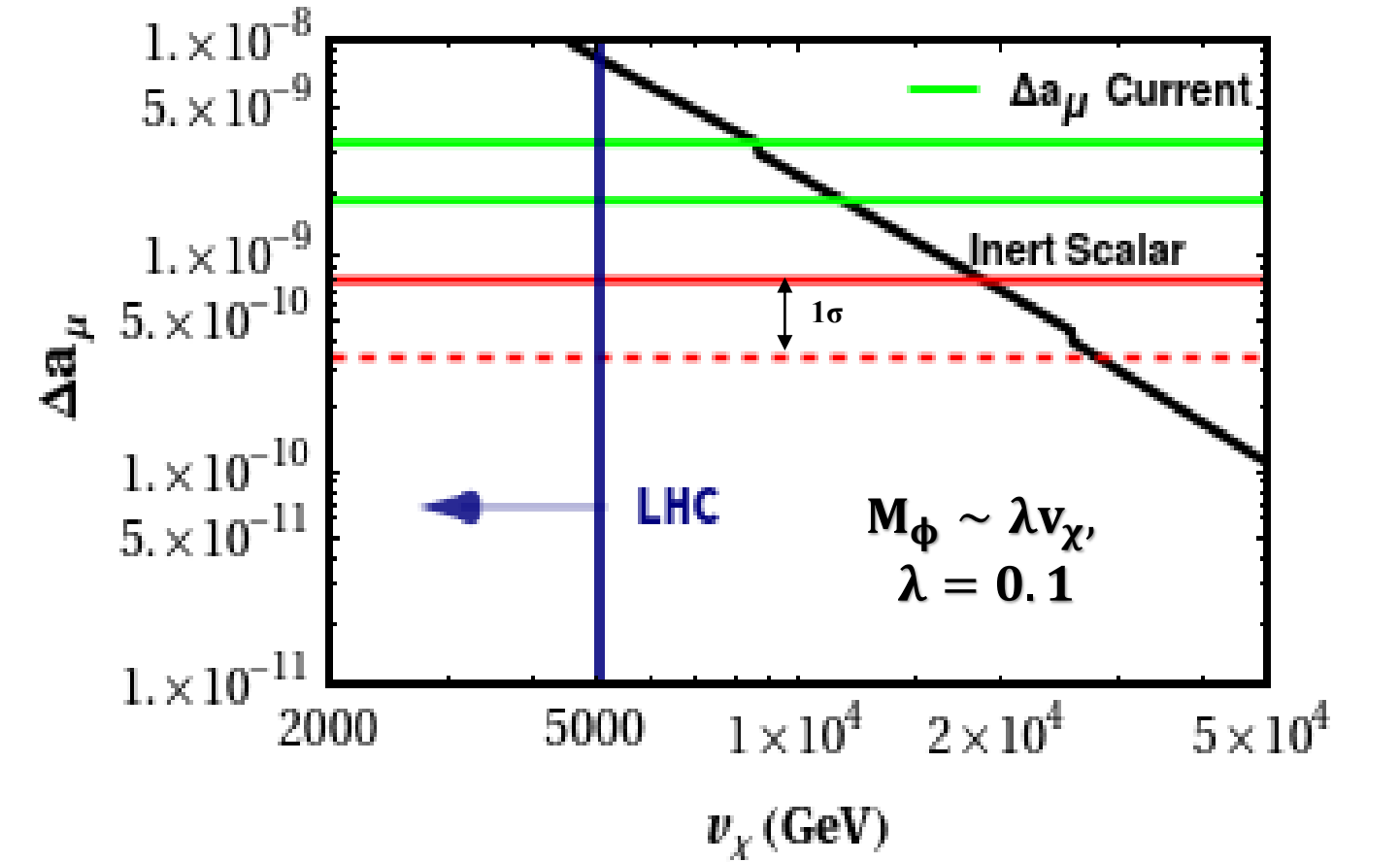


Figure 5: Overall contribution of the 3-3-1 LHN Model augmented by an inert scalar triplet ϕ . One can notice that this extended version of the 3-3-1 LHN Model now successfully accommodates the a_μ anomaly for $v_\chi \sim 10 \text{ TeV}$. The green bands are delimited by $\Delta a_\mu = (261 \pm 78) \times 10^{-11} (3.3\sigma)$. The projected 1σ bound is found by requiring $\Delta a_\mu < 78 \times 10^{-11}$ while the bound is obtained for $\Delta a_\mu < 34 \times 10^{-11}$.

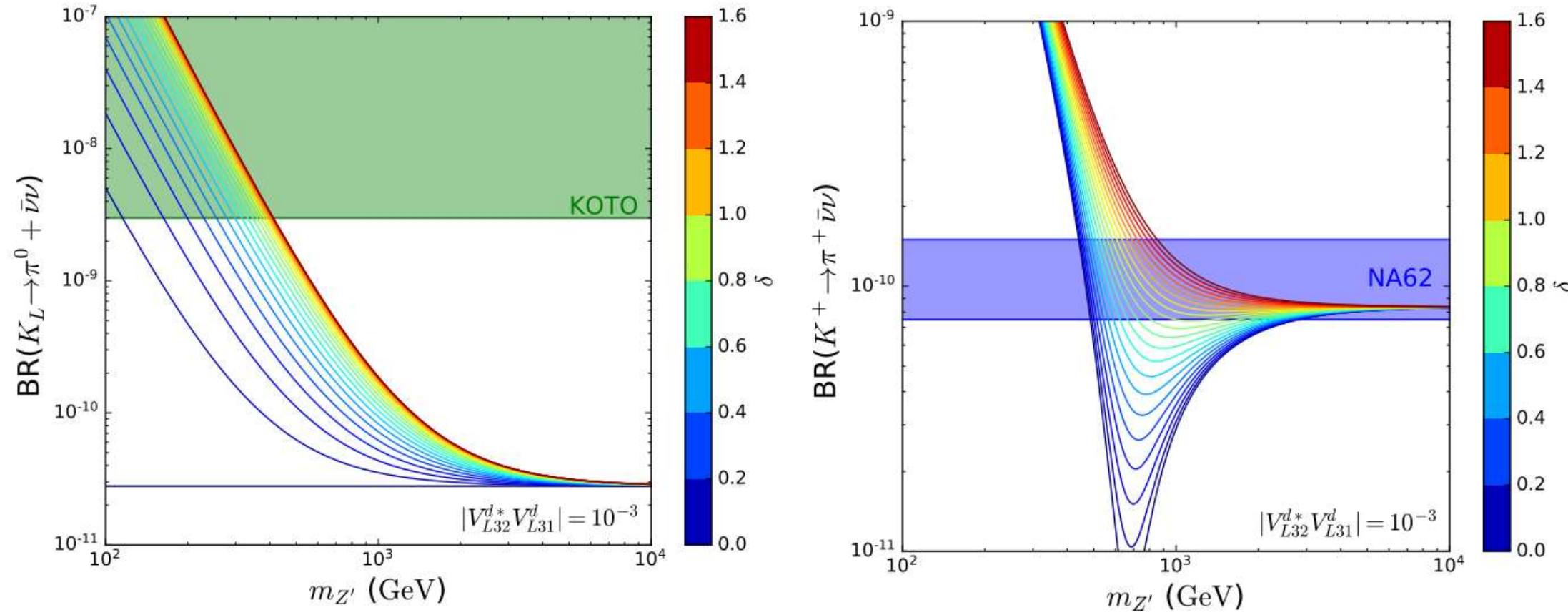
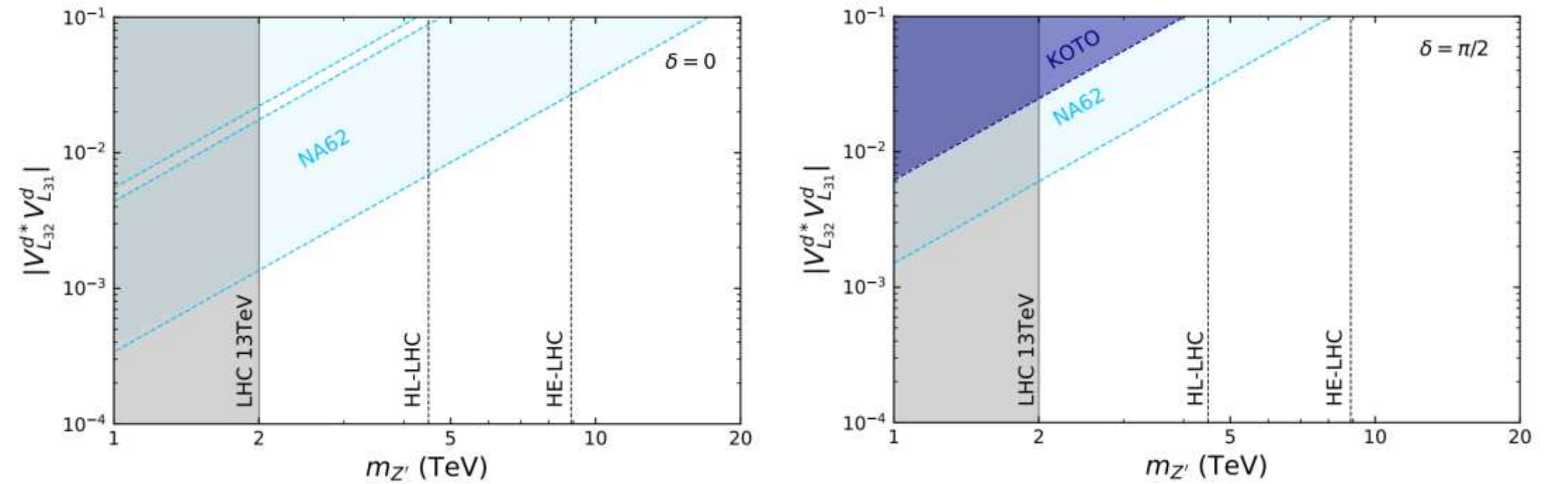


Figure 6: Branching ratio versus $m_{Z'}$ for kaon long (left panel), $BR(K_L \rightarrow \pi^0 \nu \bar{\nu})$, and for $BR(K^+ \rightarrow \pi^+ \nu \bar{\nu})$ (right panel). The green band provides the excluded region by KOTO experiment, and the blue region provides the allowed by the NA62 experiment. The colored bar represents the variation of the δ phase.

Figure 7: Excluded parameter space regions in the 331 r.h.n model, in the plane $|V_{L32}^{d*} V_{L31}^d|$ versus $m_{Z'}$, $\delta = 0$ (left panel) and $\delta = \pi/2$ (right panel). The colored regions represent the limits from NA62 (cyan), KOTO (blue) and LHC current (gray) and prospects (dashed lines).



Conclusion

— PART 4

- ✚ We concluded that none of the five models investigated here can accommodate the anomaly.
- ✚ We derived robust and complementary 1σ lower mass bounds on the masses of the new gauge bosons, namely the Z' and W' bosons.
- ✚ If the anomaly observed in the muon anomalous magnetic moment is confirmed by the $g-2$ experiment at FERMILAB these models must be extended.
- ✚ We presented a plausible extension to the 3-3-1 LHN model, which features an inert scalar triplet.
- ✚ We explored the FCNC processes mediated by a Z' gauge boson featuring both in the 3-3-1 r.h.n and 33-1LHN models.
- ✚ We computed the K^+ and K_L decay rates to missing energy, considering the extra contributions from the Z' in addition to the SM contribution.
- ✚ We performed a complementary analysis using the results from NA62, KOTO, and the LHC (current and prospects) to set bounds on the 3-3-1 r.h.n parameters.
- ✚ We found that the last result of the NA62 experiment was able to constrain a large region of the parameter space, setting lower limits on the Z' mass that can be more stringent than those from dilepton searches at the LHC.

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